

COST ACTIONS

COST Open Call Proposal

European Network on Urban Heat Resilience URBANHEAT

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Abstract

URBANHEAT – European Network on Urban Heat Resilience – is a four-year COST Action establishing a durable, inclusive, and interdisciplinary pan-European network to coordinate and advance knowledge on urban heat adaptation across climate science, urban planning, public health, and governance. Led by the University of Athens, the network unites researchers, Young Researchers and Innovators (YRI), and urban stakeholders from at least ten COST countries, with a majority from Inclusiveness Target Countries (ITC). Organised into five Working Groups covering heat metrics and monitoring, nature-based solutions, heat–health impacts, governance and policy, and dissemination and capacity building, URBANHEAT will produce harmonised methods, open-access synthesis outputs, and co-designed policy tools that no single national project or discipline can achieve alone. Key network-level deliverables include a harmonised heat stress metrics protocol, a living evidence map of nature-based solution effectiveness, a cross-national heat-health burden database, and a comparative governance typology. Networking activities include annual Training Schools targeting at least 120 YRI, Short-Term Scientific Missions, Management Committee and Working Group meetings, and ITC and YRI Conference Grants. Open Science practices — including FAIR data deposition, preprint publication, and open peer review — will be embedded as network norms from year one, and gender balance will be actively monitored and maintained across all governance structures. URBANHEAT directly supports the EU Adaptation Strategy, the EU Mission on Climate-Neutral and Smart Cities, and the European Green Deal. Its lasting legacy will be a self-sustaining European Urban Heat Resilience Community of Practice, formalised through a Community of Practice charter by month 47 to ensure continuity well beyond COST funding.

Keywords: urban heat island, climate adaptation, nature-based solutions, heat-health, urban governance, interdisciplinary network, Inclusiveness Target Countries, Young Researchers and Innovators, open science, urban resilience

SECTION 1 - SCIENCE AND TECHNOLOGY EXCELLENCE

1.1 Challenge and S&T Objectives

Urban heat is one of the most pressing and rapidly escalating environmental and public health challenges in Europe. Rising urban temperatures, intensified by the urban heat island (UHI) effect and accelerating climate change, are causing excess mortality, deepening social inequalities, straining public health systems, and undermining urban sustainability. Despite this urgency, knowledge on urban heat adaptation remains fragmented across national boundaries, disciplinary silos, and disconnected policy domains. Climate scientists, urban planners, public health researchers, social scientists, and municipal practitioners largely work in isolation, producing evidence that rarely translates into coordinated, Europe-wide adaptive strategies.

URBANHEAT will establish a durable, inclusive, and interdisciplinary pan-European research and stakeholder network to coordinate, synthesise, and advance knowledge on urban heat adaptation across four complementary domains: heat metrics and monitoring, nature-based solutions (NbS), heat–health impacts, and urban governance. The Action pursues five specific, measurable, and time-bound objectives: (1) harmonise urban heat monitoring methodologies and develop shared, interoperable heat stress metrics applicable across diverse European urban contexts by month 18; (2) systematically assess the effectiveness, co-benefits, and transferability of nature-based and built-environment solutions for urban heat mitigation, culminating in a living open-access evidence map by month 24; (3) synthesise evidence on heat-attributable health burdens with particular attention to vulnerable populations and socially deprived urban areas, producing a cross-national comparative database by month 24; (4) analyse governance frameworks, policy instruments, and institutional capacities that enable or constrain urban heat adaptation at city and national levels, delivering a comparative typology by month 30; and (5) build a sustained, open, and inclusive European community of researchers, Young Researchers and Innovators (YRI), urban practitioners, and policymakers sharing methods, tools, and findings through coordinated training, mobility, and dissemination activities throughout all four grant periods.

These objectives align directly with EU policy priorities including the EU Adaptation Strategy, the EU Mission on Climate-Neutral and Smart Cities, the European Green Deal, and the EU4Health programme. URBANHEAT will generate the cross-border, cross-disciplinary coordination that no single national project or discipline can achieve alone, directly addressing the COST mission of strengthening Europe's capacity to tackle complex societal challenges through excellent and inclusive networking.

1.2 Progress Beyond the State of the Art

The state of the art in urban heat research is advancing but remains critically fragmented. Numerous national and regional studies have characterised urban heat islands, modelled heat stress indices, and evaluated cooling interventions; however, no systematic pan-European synthesis has consolidated these efforts into shared methods or transferable knowledge. Existing initiatives — including the Copernicus Urban Atlas, the WHO Regional Office for Europe heat-health guidance, and EU-funded projects such as RESIN, URBAN-HYDREX, and CLARITY — have produced valuable partial outputs, but their temporal scope has ended and their disciplinary focus has limited cross-sector integration.

Several COST Actions have addressed adjacent topics, including CA18239 (Circular City) and CA17133 (Urbanfluxes), but none has placed urban heat adaptation at the centre of an integrated, multi-disciplinary coordination effort simultaneously spanning climate science, urban planning, public health, and governance, while maintaining a strong inclusiveness mandate. URBANHEAT advances beyond this state of the art on four fronts: (i) it establishes the first Europe-wide harmonised urban

heat monitoring protocol benchmarked against common stress thresholds; (ii) it produces a living, open-access evidence map of NbS effectiveness across European climate zones; (iii) it develops a cross-national health burden database enabling comparative epidemiological analysis; and (iv) it generates a comparative governance typology identifying best-practice institutional arrangements for heat action planning. These are genuinely novel, network-level outputs that cannot be produced by any single research group or nation and that require the sustained multi-year coordination that a COST Action uniquely enables. Together, they represent a step-change in Europe's collective capacity to understand and respond to urban heat, rather than an incremental addition to existing knowledge.

1.3 Scientific and Technological Approach

URBANHEAT is organised around five thematic Working Groups (WGs) that together span the full knowledge chain from measurement and evidence synthesis through to governance and communication. WG1 (Heat Metrics and Monitoring) will harmonise observational datasets and develop interoperable heat stress indices. WG2 (Nature-Based Solutions) will coordinate experimental and observational assessments of urban greening, blue infrastructure, and albedo interventions. WG3 (Heat and Health) will synthesise epidemiological and exposure data, co-producing a shared urban heat vulnerability mapping framework. WG4 (Governance and Policy) will conduct comparative institutional analysis and develop policy briefs co-designed with city stakeholders. WG5 (Dissemination and Capacity Building) will coordinate communication, open science outputs, training schools, and YRI mentoring across the entire network.

The networking methodology relies on four COST-funded instruments: (a) annual Management Committee (MC) meetings and plenary network conferences to maintain strategic coherence; (b) Working Group meetings (two per WG per year) to advance thematic work; (c) Short-Term Scientific Missions (STSMs) enabling early-career researchers to spend one to three months at partner institutions in different COST countries; and (d) annual Training Schools providing intensive, hands-on learning for YRI in heat monitoring, modelling, and policy analysis. ITC and YRI Conference Grants will support participation in international scientific events, maximising the network's geographic and career-stage reach. All activities will be structured under annual Work and Budget Plans agreed by the MC and approved by the COST Association.

Open Science principles will be embedded from the outset: all datasets, protocols, and synthesis reports will be published under open licences in FAIR-compliant repositories, and preprint publication will be actively encouraged as a network norm. Stakeholder co-production workshops with city practitioners, public health agencies, and urban planning departments will be integrated into WG4 activities from year one, ensuring that research outputs are demand-led and policy-relevant. The network will adopt a structured knowledge-brokerage approach, with a Science Communication Coordinator and a Grant Awarding Coordinator appointed as dedicated roles within the MC to ensure both visibility and equitable access to network opportunities.

1.4 Interdisciplinary Considerations

URBANHEAT is intrinsically and irreducibly interdisciplinary. The urban heat challenge cannot be adequately addressed by any single discipline: quantifying heat stress requires atmospheric and climate science; designing effective interventions requires urban planning and landscape architecture; understanding health consequences requires epidemiology and public health; and enabling policy change requires political science, governance studies, and social science. Each of these disciplines operates with its own methods, datasets, and publication venues, and the persistent absence of cross-disciplinary dialogue has been repeatedly identified in the literature as a primary barrier to effective urban heat adaptation across Europe.

URBANHEAT will operationalise interdisciplinarity through four structural mechanisms. First, each

Working Group will include mandatory cross-disciplinary membership, with WG leaders drawn from different fields to prevent any single discipline from dominating the agenda. Second, joint WG task forces will be constituted for crosscutting deliverables — specifically, the urban heat vulnerability mapping framework (WG1 × WG3) and the NbS governance toolkit (WG2 × WG4) — ensuring that integrated outputs are produced collaboratively rather than by post-hoc aggregation. Third, Training Schools will be designed around problem-based learning scenarios that require participants from different disciplines to collaborate on shared urban case studies, building habits of interdisciplinary reasoning in the next generation of researchers. Fourth, the network will actively recruit researchers from the social sciences and humanities, engineering, and medicine alongside natural scientists, ensuring that the MC itself reflects genuine disciplinary diversity rather than a natural-science majority with token representation from other fields. By bridging these communities within a single durable network, URBANHEAT will catalyse methodological cross-fertilisation and produce integrative knowledge that advances the field well beyond the sum of its disciplinary parts. Where disciplinary tensions arise — for example, between quantitative epidemiological standards and qualitative governance research — these will be treated as productive intellectual encounters rather than obstacles, and WG5 will provide dedicated guidance on interdisciplinary co-authorship and joint methods reporting.

SECTION 2 - IMPACT

2.1 Expected Impact and Benefits

The primary pathway to scientific impact runs through the production of shared, interoperable methods and synthesis outputs that individual research groups can build upon and extend. The harmonised heat stress metrics protocol (WG1, month 18), the living evidence map on nature-based solutions (WG2, month 24), the cross-national health burden database (WG3, month 24), and the governance typology (WG4, month 30) will collectively constitute a European knowledge infrastructure for urban heat adaptation — one that will catalyse and inform national research programmes, city adaptation plans, and EU policy instruments for years beyond the Action's duration. The network targets at least 20 peer-reviewed open-access publications by month 48, contributing directly to the scientific literature across climate science, public health, urban planning, and governance journals.

The pathway to policy impact is structured through WG4's co-design workshops with city authorities, national adaptation focal points, and public health agencies. Policy briefs produced jointly with stakeholders will target EU institutions (DG CLIMA, DG SANTE, the EU Mission on Cities), national ministries, and municipal governments, using plain-language summaries and actionable recommendations. The network will actively seek engagement with the European Environment Agency, the WHO Regional Office for Europe, and ICLEI – Local Governments for Sustainability as key multiplier organisations capable of amplifying policy uptake beyond the immediate network. At least 30 municipal or public health stakeholder organisations will be formally engaged by month 36.

Societal impact will be generated by improving the quality, equity, and evidence base of urban heat governance, with particular emphasis on protecting vulnerable groups including elderly populations, outdoor workers, and residents of socially deprived neighbourhoods. URBANHEAT's capacity-building activities — targeting at least 120 YRI across four Training Schools — will create a generation of early-career researchers trained in interdisciplinary heat adaptation science, multiplying the network's long-term influence far beyond its four-year operational period. By year four, the network will establish a self-sustaining European Urban Heat Resilience Community of Practice, formalised through a Community of Practice charter (D5.3), providing institutional continuity and a membership structure that can attract new participants and resources after COST funding ends.

2.2 Dissemination and Stakeholder Engagement

URBANHEAT will implement a layered dissemination strategy addressing scientific, policy, and public audiences in a coordinated and sequenced manner throughout the Action's lifetime.

For scientific audiences, the network will publish at least 20 peer-reviewed papers in open-access journals by month 48, contribute organised sessions at major international conferences (EGU, EMS, ISEE, AESOP), and maintain a curated, publicly accessible repository of shared datasets, protocols, and synthesis outputs. All outputs will comply with FAIR data principles and will be deposited in Zenodo under Creative Commons licences, ensuring long-term discoverability and reuse.

For policy and practitioner audiences, WG4 will produce six policy briefs co-authored with city stakeholders, a comparative heat action planning guide, and annual network newsletters distributed to more than 200 municipal and regional authorities across Europe. A dedicated URBANHEAT web portal, launched by month 6 (D5.1), will serve as the central hub for open-access resources, stakeholder engagement, and community membership, and will remain operational beyond the grant period through a hosting agreement with the University of Athens.

The network will take specific and proactive measures to ensure equitable participation in

dissemination. ITC Conference Grants will cover travel to international scientific meetings for researchers based in Inclusiveness Target Countries, and at least one Training School per grant period will be hosted in an ITC country to reduce access barriers. YRI Conference Grants will further support early-career researchers in presenting their work at major disciplinary venues.

Gender-disaggregated participation data will be monitored and publicly reported annually by WG5 (D5.2). If gender imbalance is detected in any WG or governance body — defined as any single gender exceeding 60% of membership — targeted corrective measures will be applied, including directed outreach and adjusted nomination procedures. Open Science practices — including preprint deposition, open peer review, and proactive data sharing — will be promoted as network norms through dedicated guidance developed by WG5 and communicated to all network members from year one.

2.3 Summary of Impact Indicators

Indicator	Target	Measurement
Peer-reviewed open-access publications	At least 20 by month 48	Tracked via DOI register and network publication list
Policy briefs disseminated to municipal/national authorities	At least 6 by month 48	Tracked by WG4 and Science Communication Coordinator
Short-Term Scientific Missions completed	At least 24 over 4 grant periods	COST e-COST reporting system
Researchers trained at Training Schools	At least 120 YRI over 4 grant periods	Training School attendance registers
ITC country representation in MC and WGs	At least 50% of participating COST countries from ITC list	Annual MC membership review
YRI share in network activities	At least 40% of active participants classified as YRI	Annual network survey and WG reports
City/stakeholder organisations engaged	At least 30 municipal or public health stakeholders by month 36	Stakeholder register maintained by WG4
Open datasets and protocols deposited	At least 5 FAIR-compliant datasets by month 48	Zenodo repository DOI records

SECTION 3 - IMPLEMENTATION AND NETWORKING

3.1 Working Groups and Activities

WP	Title	Lead	PM	Duration
WP1	WG1 – Heat Metrics and Monitoring	University of Athens, Greece	0	M1-M48
WP2	WG2 – Nature-Based Solutions for Urban Cooling	TU Wien, Austria	0	M1-M48
WP3	WG3 – Heat and Health	University of Tartu, Estonia	0	M1-M48
WP4	WG4 – Governance, Policy and Stakeholder Engagement	University of Belgrade, Serbia	0	M1-M48
WP5	WG5 – Dissemination, Capacity Building and Open Science	University of Ljubljana, Slovenia	0	M1-M48

WP1: WG1 – Heat Metrics and Monitoring

Lead: University of Athens, Greece | **Person-months:** 0 | **Duration:** M1 - M48

WG1 will harmonise urban heat monitoring methodologies across the network, develop interoperable heat stress indices adapted to diverse European urban morphologies and climate zones, and produce an open-access protocol repository accessible to all COST member countries. Activities include two WG meetings per year, one STSM programme per grant period focused on observational data exchange between national monitoring networks, and co-authorship of peer-reviewed synthesis papers. WG1 outputs feed directly into WG2 (providing climate-zone classifications for NbS transferability assessments), WG3 (supplying harmonised exposure data for health burden analysis), and WG4 (informing monitoring requirements within governance frameworks). The WG will establish a data standards task force in month one to resolve inter-country harmonisation challenges early and to define minimum common data standards that can accommodate national legislative differences through metadata overlays.

T1.1: Inventory of national monitoring practices - Compile and systematically compare existing urban heat monitoring datasets, station networks, and heat stress index definitions across participating COST countries, identifying gaps, overlaps, and opportunities for harmonisation. The inventory will be published as D1.1 (month 9) and will form the evidence base for T1.2.

T1.2: Harmonised metrics protocol - Develop and publish an agreed, interoperable heat stress metrics protocol benchmarked against WHO thresholds and Copernicus Climate Change Service standards, validated through cross-network pilot applications in at least five cities spanning different European climate zones. The protocol will be deposited as an open-access document in Zenodo (D1.2, month 18).

T1.3: Open-access data repository - Establish a FAIR-compliant shared dataset repository linking WG1 harmonised outputs to the analytical needs of WG2, WG3, and WG4, with a data management plan reviewed annually by the MC. The repository will adopt persistent identifiers and documented metadata schemas to ensure long-term accessibility beyond the Action's lifetime.

WP2: WG2 – Nature-Based Solutions for Urban Cooling

Lead: TU Wien, Austria | **Person-months:** 0 | **Duration:** M1 - M48

WG2 will systematically review and assess the cooling effectiveness, co-benefits, and transferability of nature-based solutions (NbS) — including urban greening, blue infrastructure, cool and permeable pavements, and green roofs — across European climate zones. The WG will produce a living evidence map updated annually and will organise at least one STSM programme per grant period enabling cross-site comparative field assessments. WG2 works in close liaison with WG1 (climate-zone data), WG3 (health and equity co-benefits of NbS), and WG4 (governance conditions enabling NbS deployment at scale), ensuring that NbS assessments are embedded within broader adaptation planning contexts rather than evaluated in technical isolation.

T2.1: Living evidence map of NbS effectiveness - Conduct a systematic review and living synthesis of quantitative studies on the cooling performance of nature-based solutions across European urban contexts, following established systematic mapping protocols. The first version (v1.0) will be published as D2.1 by month 24 and updated annually thereafter, with all underlying data deposited in an open repository.

T2.2: Transferability assessment framework - Develop a decision-support framework identifying the climatic, morphological, and socio-economic conditions under which specific NbS are transferable across European climate zones and urban typologies. The framework will be co-developed with WG4 to ensure that governance and institutional preconditions are integrated alongside technical criteria, and will be published as D2.2 (month 36).

T2.3: Co-benefits and trade-offs analysis - Assess biodiversity, water management, social equity, and air quality co-benefits of urban cooling interventions, drawing on WG3 health data and WG4 governance insights. The analysis will explicitly address potential trade-offs — such as water demand of urban greening in drought-prone cities — to ensure that recommendations are context-sensitive and operationally credible for city practitioners.

WP3: WG3 – Heat and Health

Lead: University of Tartu, Estonia | Person-months: 0 | Duration: M1 - M48

WG3 will synthesise epidemiological and exposure data on heat-attributable health burdens across COST member countries, with particular emphasis on vulnerable populations — including the elderly, outdoor workers, and residents of socially deprived urban areas — and on the intersections between heat exposure, pre-existing conditions, and adaptive capacity. The WG will develop a cross-national comparative health burden database and a shared vulnerability mapping methodology co-designed with public health stakeholders and aligned with WG1 exposure metrics. WG3 outputs will directly inform WG4 policy briefs and the heat action planning guide, ensuring that health evidence is translated into governance-relevant recommendations.

T3.1: Cross-national heat-health burden database - Compile and harmonise heat-attributable mortality and morbidity data from national health registries and published epidemiological studies across at least 15 COST countries, applying standardised attribution methods and producing a publicly accessible database (D3.1, month 24) with full documentation of data sources, inclusion criteria, and methodological limitations.

T3.2: Vulnerability mapping methodology - Develop a standardised, transferable urban heat vulnerability mapping framework integrating socio-demographic, exposure, and adaptive capacity indicators. The framework will be co-designed with WG1 (exposure metrics) and WG4 (governance and equity dimensions), tested in at least three pilot cities across different national contexts, and published as a documented methodological guide (D3.2, month 32).

T3.3: Policy-relevant health evidence synthesis - Produce synthesis reports and policy briefs on heat-health evidence for European public health agencies and national adaptation authorities, co-authored with WG4. Outputs will translate epidemiological findings into actionable public health recommendations, including thresholds for early warning systems and priority intervention groups, targeting audiences at WHO Regional Office for Europe, DG SANTE, and national ministries of health.

WP4: WG4 – Governance, Policy and Stakeholder Engagement

Lead: University of Belgrade, Serbia | Person-months: 0 | Duration: M1 - M48

WG4 will analyse urban heat governance frameworks and policy instruments across Europe, conduct annual co-design workshops with city authorities and public health agencies, and produce a comparative governance typology and a suite of co-authored policy briefs. It will serve as the primary interface between the network and external policy stakeholders, maintaining the network's stakeholder register and coordinating engagement with EU institutions, national ministries, and municipal authorities. WG4 co-leads the vulnerability mapping framework with WG3, co-develops the NbS transferability framework with WG2, and ensures that all WG outputs are translated into governance-relevant language accessible to non-specialist audiences. Stakeholder relationships will be embedded at the organisational level, with at least three institutional contact points per city partner, to maintain continuity across electoral and administrative cycles.

T4.1: Comparative governance analysis - Map and compare institutional arrangements, legislative frameworks, and policy instruments for urban heat adaptation across at least 20 COST countries, using a structured comparative analysis protocol developed in month one. The analysis will identify enabling conditions, barriers, and governance gaps, and will be published as a scientific report (D4.1, month 30) that forms the empirical basis for the policy briefs.

T4.2: Co-design workshops with city stakeholders - Organise annual co-design workshops bringing together municipal authorities, urban planners, and public health agencies to co-produce demand-led research questions, review emerging findings, and validate draft policy outputs. Workshops will be structured around specific deliverables at each stage (governance analysis, policy briefs, planning guide) to ensure that stakeholder input is substantive rather

than consultative.

T4.3: Policy briefs and heat action planning guide - Produce six policy briefs (D4.2, month 42) and one comparative heat action planning guide (D4.3, month 44) co-authored with stakeholder partners. Policy briefs will target specific audiences — EU institutions (DG CLIMA, DG SANTE, EU Mission on Cities), national ministries, and city governments — with differentiated language and formats. The planning guide will provide a step-by-step decision-support resource for cities developing or revising heat action plans.

WP5: WG5 – Dissemination, Capacity Building and Open Science

Lead: University of Ljubljana, Slovenia | Person-months: 0 | Duration: M1 - M48

WG5 will coordinate all dissemination, communication, training, and capacity-building activities across the network, including the annual Training Schools, the YRI mentoring programme, the URBANHEAT web portal, the Open Science implementation plan, and the network's gender balance and YRI inclusion monitoring. WG5 is responsible for ensuring that the network's outputs reach the widest possible scientific, policy, and public audiences, and that the network itself is equitable, inclusive, and transparent in its internal processes. WG5 will report annually to the MC on gender-disaggregated and career-stage participation data across all WGs and governance bodies, recommending corrective measures wherever targets are not met. WG5 also leads the preparation of the final network synthesis report and the Community of Practice charter that will provide institutional continuity beyond COST funding.

T5.1: Annual Training Schools - Organise one Training School per grant period, hosted on a rotating basis across COST countries with preference for ITC host institutions. Each school will provide intensive, interdisciplinary, problem-based training for at least 30 YRI in urban heat monitoring, NbS assessment, epidemiological methods, and policy analysis. A structured evaluation will be conducted after each school to improve subsequent editions, and anonymised evaluation reports will be shared with the MC.

T5.2: URBANHEAT web portal and open-access repository - Develop and maintain the network's web portal (launched by month 6 as D5.1) and a linked Zenodo community space hosting all open-access outputs, datasets, protocols, and tools. The portal will include a public membership directory, an events calendar, a news feed, and a searchable resource library. Hosting and maintenance arrangements with the University of Athens will be confirmed by month 6 to ensure continuity beyond the grant period.

T5.3: Gender balance and YRI inclusion monitoring - Implement annual monitoring of gender and career-stage (YRI) participation across all WGs, governance bodies, Training Schools, STSMs, and grant allocations. Annual reports (D5.2) will be submitted to the MC with quantitative data disaggregated by gender, career stage, and country grouping (ITC vs. non-ITC). Where targets — specifically, no gender exceeding 60% of any WG leadership, and at least 40% YRI across active participants — are not met, the MC will be presented with concrete corrective options for adoption.

3.2 Network of Proposers

Organisation	Country	Type	Role	Description
University of Athens	Greece	University	Coordinator	Main Proposer and Action Chair institution; leading WG1 and overall network coordination
TU Wien	Austria	University	Partner	WG2 lead; nature-based solutions and urban climatology expertise
University of Belgrade	Serbia	University	Partner	WG4 lead; governance and policy analysis; ITC anchor institution
University of Tartu	Estonia	University	Partner	WG3 lead; heat-health epidemiology; ITC anchor institution
University of Ljubljana	Slovenia	University	Partner	WG5 lead; dissemination, training and open science coordination;

				ITC anchor institution
Warsaw University of Technology	Poland	University	Partner	Urban planning and heat modelling; ITC anchor institution
Czech Technical University in Prague	Czech Republic	University	Partner	Remote sensing and urban heat island monitoring; ITC anchor institution
University of Lisbon	Portugal	University	Partner	Mediterranean urban heat and nature-based solutions expertise
KU Leuven	Belgium	University	Partner	Public health and epidemiology expertise; YRI programme support
University of Copenhagen	Denmark	University	Partner	Urban governance and climate adaptation policy expertise
Sofia University St Kliment Ohridski	Bulgaria	University	Partner	Vulnerability mapping and social science; ITC anchor institution

URBANHEAT brings together a geographically diverse network spanning at least ten COST countries, with a majority of participating institutions located in Inclusiveness Target Countries (Serbia, Estonia, Slovenia, Poland, Czech Republic, and Bulgaria). This composition directly fulfils the COST Excellence and Inclusiveness Policy requirement that at least 50% of represented COST Full or Cooperating Members be drawn from the ITC list. Critically, the network is designed so that ITC institutions hold genuine scientific leadership — leading four of the five Working Groups — rather than serving merely as participation statistics. This ensures authentic capacity building and bidirectional knowledge transfer rather than a one-directional flow of expertise from Western Europe.

The consortium deliberately combines research-intensive institutions from Western Europe (TU Wien, KU Leuven, University of Copenhagen, University of Lisbon) with emerging and growing research communities in Central, Eastern, and Southern Europe. This diversity is scientifically productive as well as strategically appropriate: Southern and Eastern European cities face some of the most severe urban heat burdens in Europe, and the researchers working in those contexts bring indispensable empirical knowledge and place-based expertise to the network.

Young Researchers and Innovators are central to the network's design and governance, not an afterthought. At least 40% of active participants across all Working Groups will be YRI, supported through dedicated STSMs, Training School places, YRI Conference Grants, and a structured cross-country mentoring programme pairing early-career researchers with senior network members. The inclusion of YRI among the initial Secondary Proposers in the Network of Proposers signals this commitment from the outset.

Gender balance will be monitored annually and actively promoted. No gender will be permitted to exceed 60% of any Working Group's leadership positions, and targeted outreach will be directed at underrepresented researchers throughout the network's lifetime. The network's open, bottom-up character — consistent with COST's inclusive networking philosophy — means that new participants from any COST member country will be welcomed at any point during the four-year lifetime, ensuring continuous broadening of the network's geographic, disciplinary, and demographic coverage. [REVIEWER NOTE: The proposal should verify at submission that the initial Network of

Proposers document formally lists the Secondary Proposers from ITC countries as required by COST Open Call eligibility rules, and that the ITC country count meets the minimum threshold specified in the current COST Vademecum.]

3.3 Milestones

ID	Title	WP	Due	Verification
MS1	Network launched, MC constituted, WG leaders appointed	All WGs	M3	MC constitution report submitted to COST Association
MS2	Harmonised heat stress metrics protocol published	WG1	M18	Open-access protocol document deposited in Zenodo with DOI
MS3	Living evidence map on nature-based solutions (v1.0) published	WG2	M24	Peer-reviewed publication submitted and evidence map accessible online
MS4	Cross-national heat-health burden database (v1.0) complete	WG3	M24	Database accessible in open repository with data management plan
MS5	Comparative governance typology report completed	WG4	M30	Governance report published on URBANHEAT portal
MS6	First Training School successfully delivered	WG5	M12	Training School report and attendance register submitted to MC
MS7	All major deliverables completed; network sustainability plan adopted	All WGs	M46	Final synthesis report approved by MC; Community of Practice charter adopted

3.4 Deliverables

ID	Title	WP	Due	Type	Dissemination
D1.1	Urban heat monitoring inventory report	WG1	M9	Report	Public
D1.2	Harmonised heat stress metrics protocol	WG1	M18	Protocol document / open dataset	Public
D2.1	Living evidence map – nature-based solutions for urban cooling (v1.0)	WG2	M24	Synthesis report / online database	Public
D2.2	NbS transferability decision-support framework	WG2	M36	Guidance document	Public
D3.1	Cross-national heat-health burden database	WG3	M24	Open dataset	Public
D3.2	Urban heat vulnerability mapping methodology	WG3	M32	Technical report / methodological guide	Public
D4.1	Comparative	WG4	M30	Scientific report	Public

	urban heat governance typology				
D4.2	Set of six policy briefs for EU institutions, national and city authorities	WG4	M42	Policy briefs	Public
D4.3	Comparative heat action planning guide	WG4	M44	Practitioner guide	Public
D5.1	URBANHEAT web portal (launch version)	WG5	M6	Website / digital platform	Public
D5.2	Annual gender balance and YRI participation reports (x4)	WG5	M48	Monitoring report	Restricted to MC
D5.3	Final network synthesis report and Community of Practice charter	All WGs	M47	Synthesis report	Public

3.6 Risk Management

ID	Risk	Likelihood	Impact	Mitigation
R1	High coordination load on Action Chair and WG leaders given the network's disciplinary and geographic breadth may cause management bottlenecks.	Medium	High	Appoint a dedicated Science Communication Coordinator and Grant Awarding Coordinator from month one; distribute governance responsibilities across WG leaders; use shared online project management tools.
R2	Stakeholder continuity risk: city authority contacts and public health agency representatives may change due to elections or institutional restructuring, disrupting WG4 co-design processes.	Medium	Medium	Maintain a diversified stakeholder register with at least three institutional contacts per city; embed relationships at the organisational rather than individual level; document co-design outputs formally.
R3	Uneven participation from ITC countries, particularly if researchers from less research-intensive institutions face barriers to attending network activities due to workload or	Medium	High	Prioritise ITC-hosted meetings and Training Schools; proactively allocate STSM and conference grant budgets to ITC participants; assign ITC country mentors within the network.

	national funding constraints.			
R4	Data harmonisation difficulties across countries with different national monitoring standards, health registry formats, and legislative frameworks may delay WG1 and WG3 deliverables.	Medium	Medium	Establish a data standards task force in WG1 from month one; adopt minimum common data standards early; allow national metadata overlays to accommodate legislative differences.
R5	YRI retention risk: early-career researchers who are key network contributors may leave academia or relocate during the four-year period, disrupting WG contributions.	Low	Medium	Build redundancy into WG membership by training multiple YRI per WG; use the mentoring programme to develop YRI leadership capacity; acknowledge YRI contributions formally in all publications and reports.